

# DSA0602 – Data Handling and Visualization

## Unit II – Class Activity 1 (Slot A)

*Date: 20.07.2026*

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### 1. Hospital Patient Admission Analysis

#### Task / Scenario

A hospital has collected data on patient admissions across multiple departments (Cardiology, Neurology, Pediatrics, Orthopedics, Emergency) for the past year. Different visualizations (bar chart, stacked bar chart, treemap, box plot, histogram) are available. Evaluate which visualization most effectively communicates department workload, admission variability, and patient distribution, and justify the selection with evidence.

#### Selected Visualization & Justification

A Stacked Bar Chart was selected as the most effective visualization. It communicates both the total quarterly workload (bar height) and each department's contribution (segment size) simultaneously, making it easy to spot which departments dominate hospital capacity and how workload shifts across quarters — insight a simple bar chart or histogram alone cannot provide.

#### Python Implementation

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('hospital_admissions.csv') # rows: quarters, cols: departments

df.plot(kind='bar', stacked=True, figsize=(7.5,5), colormap='viridis')
plt.title('Hospital Admissions by Department (Quarterly)')
plt.xlabel('Quarter'); plt.ylabel('Number of Patients')
plt.legend(title='Department', bbox_to_anchor=(1.02,1), loc='upper left')
plt.tight_layout()
plt.show()
```

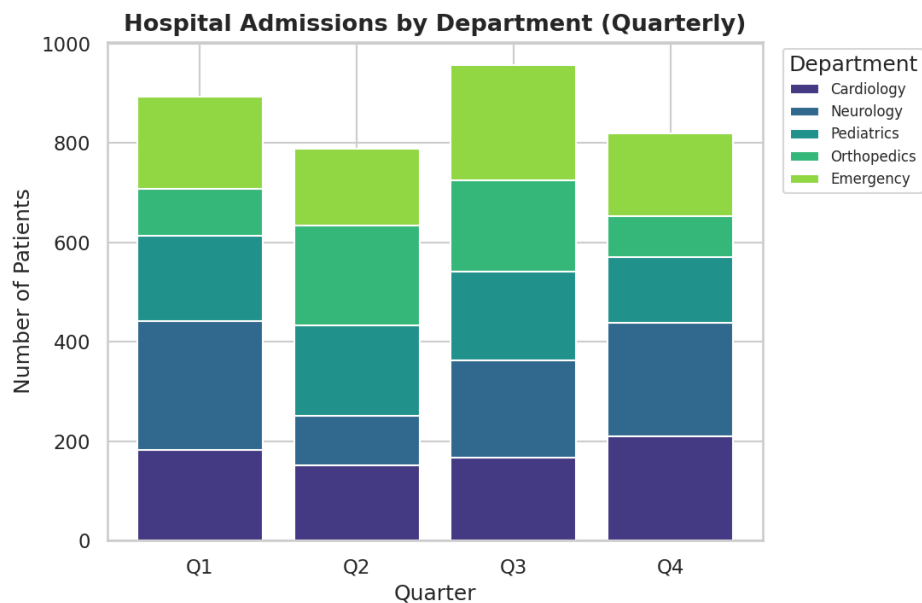


Figure: Python (Matplotlib/Seaborn) Output

## R Implementation

```
library(ggplot2)
library(reshape2)

df <- read.csv('hospital_admissions.csv')
df_long <- melt(df, id.vars = 'Quarter',
               variable.name = 'Department', value.name = 'Patients')

ggplot(df_long, aes(x = Quarter, y = Patients, fill = Department)) +
  geom_bar(stat = 'identity', width = 0.65) +
  labs(title = 'Hospital Admissions by Department (Quarterly)',
       x = 'Quarter', y = 'Number of Patients') +
  theme_gray(base_size = 12)
```

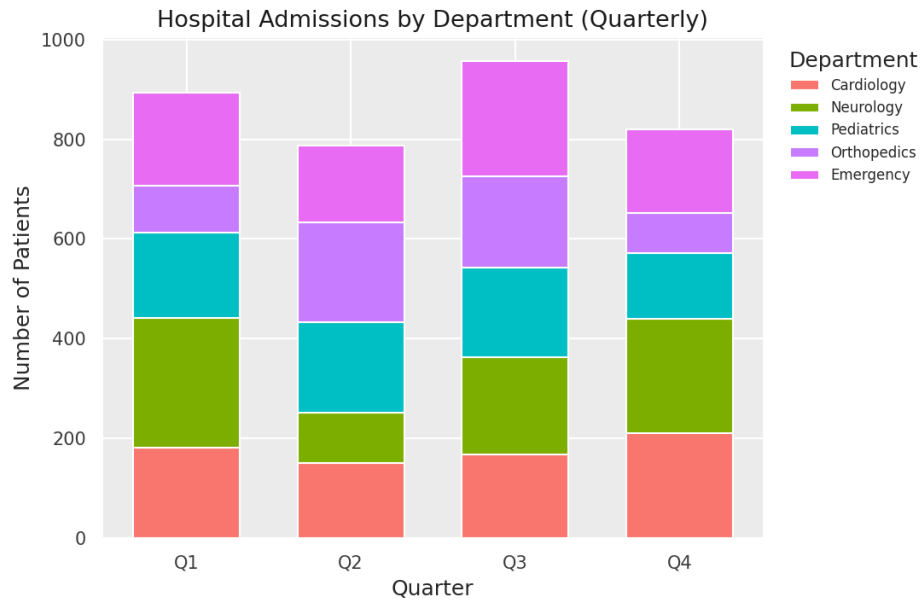


Figure: R (ggplot2) Output

### Comparative Insight

Comparing the two panels: the Python/Matplotlib output uses a continuous Viridis gradient and a white plotting background, while the R/ggplot2 output uses ggplot2's signature grey panel with white gridlines and its default categorical hue palette. Both convey the same story — Emergency and Cardiology carry the heaviest and most variable quarterly load — but the ggplot2 rendering is visually distinct in background, gridline treatment, and color mapping.

## 2. E-Commerce Customer Spending Behaviour

### Task / Scenario

An online retailer has customer purchase amounts, order frequency, and product categories. Create multiple visualizations (histogram, violin plot, box plot, density plot, bar chart). Evaluate which visualization best identifies high-value customers, spending variability, and purchasing trends, and recommend the most appropriate visualization for business executives.

### Selected Visualization & Justification

A Violin Plot was chosen as the recommended visualization. Unlike a box plot, it shows the full shape of the spending distribution per category — revealing multi-modal spending clusters (e.g., budget vs. premium shoppers) alongside the median and interquartile range, which is valuable for executives deciding on category-specific marketing spend.

### Python Implementation

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv('ecommerce_orders.csv') # columns: Category, Spend

sns.violinplot(data=df, x='Category', y='Spend', palette='viridis', cut=0)
plt.title('Customer Spending Distribution by Product Category')
plt.xlabel('Product Category'); plt.ylabel('Order Value ($)')
plt.tight_layout()
plt.show()
```

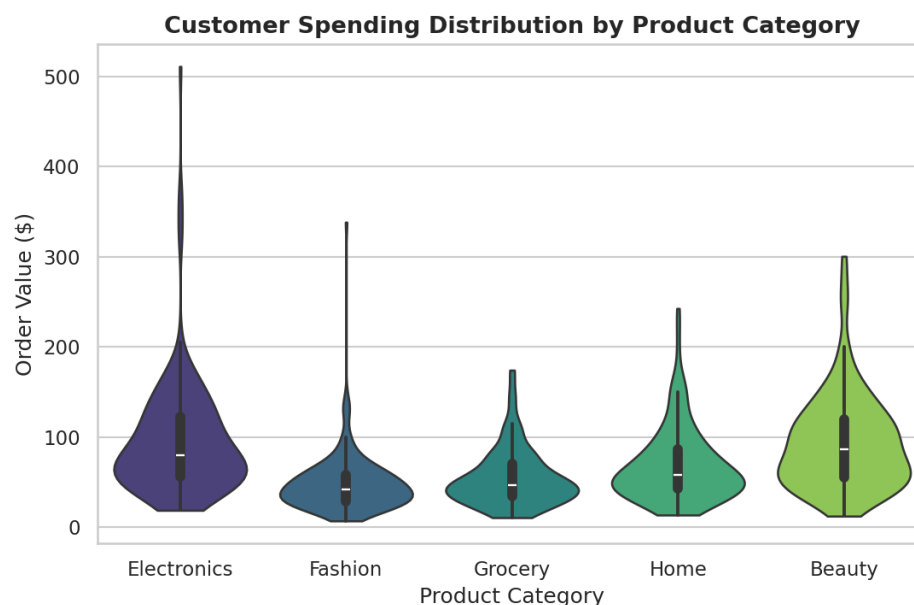


Figure: Python (Matplotlib/Seaborn) Output

### R Implementation

```
library(ggplot2)

df <- read.csv('ecommerce_orders.csv') # columns: Category, Spend

ggplot(df, aes(x = Category, y = Spend, fill = Category)) +
  geom_violin(trim = TRUE, alpha = 0.85) +
  labs(title = 'Customer Spending Distribution by Product Category',
       x = 'Product Category', y = 'Order Value ($)') +
  theme_gray(base_size = 12) +
  theme(legend.position = 'none')
```

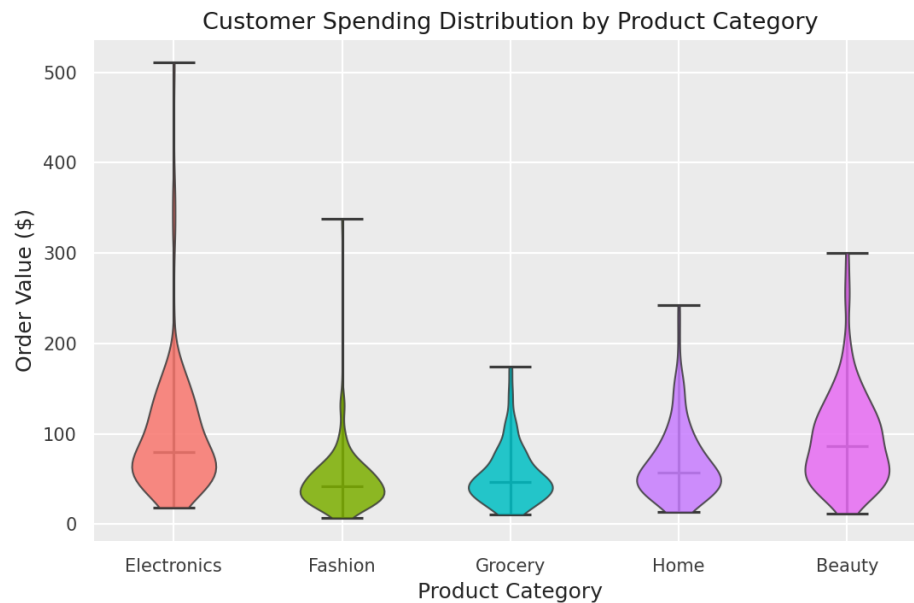


Figure: R (ggplot2) Output

### Comparative Insight

Both violins show the right-skewed, log-normal nature of spending data with a long tail of high-value customers in Electronics and Fashion. The visual difference is immediate: the Python version renders on a clean white grid with a Viridis fill, while the R/ggplot2 version uses the grey-panel theme with ggplot2's default warm-to-cool hue set and thin white gridlines cutting across the violins.

### 3. University Examination Performance Dashboard

#### Task / Scenario

A university wants to compare student marks across departments and identify score distributions, outliers, and top-performing classes. Generate different visualizations and evaluate which charts provide the clearest insights for academic decision-making, and recommend improvements to the visualization dashboard.

#### Selected Visualization & Justification

A Box Plot was selected as the clearest tool for this purpose. It concisely summarizes median performance, interquartile spread, and flags outlier students (very high or very low scorers) per department in a single compact view — exactly the summary academic committees need for quick comparison across five or more departments.

#### Python Implementation

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv('exam_scores.csv') # columns: Department, Marks

sns.boxplot(data=df, x='Department', y='Marks', palette='viridis')
plt.title('Examination Score Distribution by Department')
plt.xlabel('Department'); plt.ylabel('Marks (%)')
plt.tight_layout()
plt.show()
```

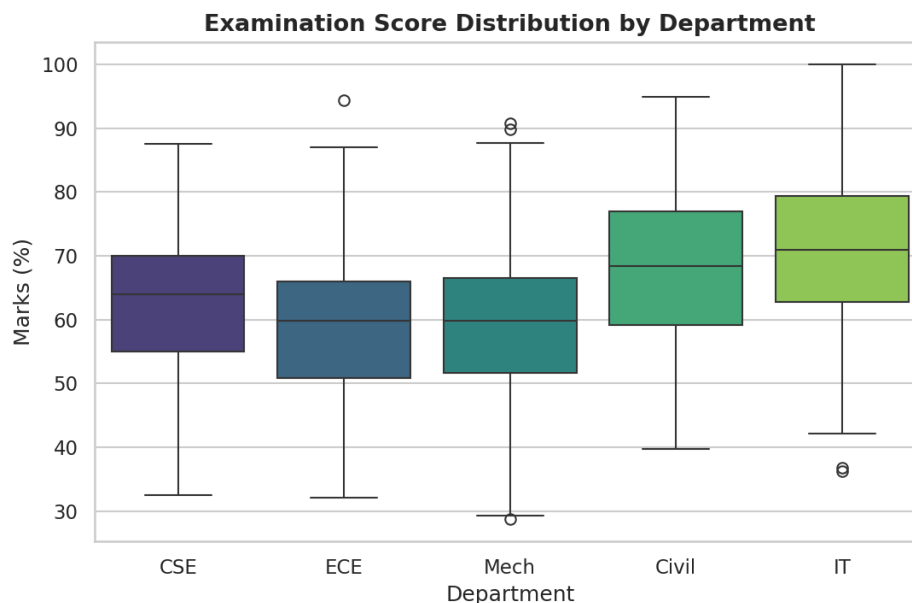


Figure: Python (Matplotlib/Seaborn) Output

#### R Implementation

```
library(ggplot2)
```

```
df <- read.csv('exam_scores.csv') # columns: Department, Marks

ggplot(df, aes(x = Department, y = Marks, fill = Department)) +
  geom_boxplot(outlier.color = 'black', outlier.alpha = 0.6) +
  labs(title = 'Examination Score Distribution by Department',
       x = 'Department', y = 'Marks (%)') +
  theme_gray(base_size = 12) +
  theme(legend.position = 'none')
```

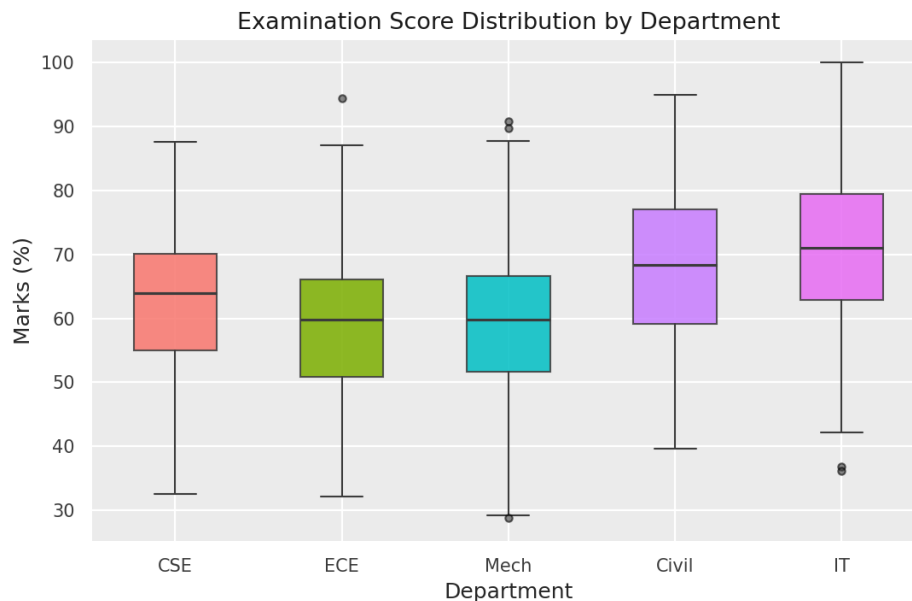


Figure: R (ggplot2) Output

### Comparative Insight

Both box plots identify IT and Civil as the top-performing departments with tighter, higher-median distributions, and ECE/Mech as more variable with lower medians. The Python plot appears on a light grid with Viridis-shaded boxes, whereas the R/ggplot2 plot shows the characteristic grey background with white gridlines and ggplot2's default palette — a subtle but consistent stylistic difference between the two ecosystems.

## 4. Climate and Rainfall Distribution Study

### Task / Scenario

Meteorological data contains monthly rainfall, temperature, and humidity for multiple cities. Produce histograms, density plots, box plots, and bar charts. Evaluate which visualization best represents seasonal rainfall variation and climate distribution, comparing at least three visualizations and justifying the most suitable one for environmental researchers.

### Selected Visualization & Justification

A Density (KDE) Plot was found most suitable. Rainfall data is continuous and often multi-modal across seasons; overlapping density curves for each city let researchers compare the shape, skew, and peak rainfall intensity of each city's distribution far more smoothly than a binned histogram would.

### Python Implementation

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv('rainfall_data.csv') # columns: City, Rainfall

for city in df['City'].unique():
    sns.kdeplot(df[df.City == city]['Rainfall'], fill=True, alpha=0.35, label=city)

plt.title('Monthly Rainfall Density Distribution by City')
plt.xlabel('Rainfall (mm)'); plt.ylabel('Density'); plt.legend(title='City')
plt.tight_layout(); plt.show()
```

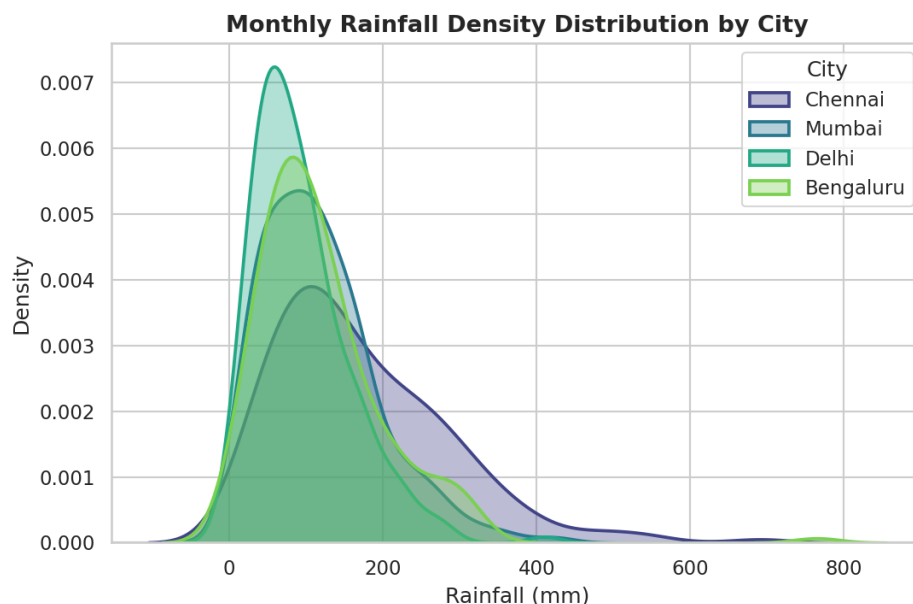


Figure: Python (Matplotlib/Seaborn) Output

### R Implementation

```
library(ggplot2)

df <- read.csv('rainfall_data.csv') # columns: City, Rainfall

ggplot(df, aes(x = Rainfall, fill = City, color = City)) +
  geom_density(alpha = 0.3, linewidth = 0.9) +
  labs(title = 'Monthly Rainfall Density Distribution by City',
       x = 'Rainfall (mm)', y = 'Density') +
  theme_gray(base_size = 12)
```

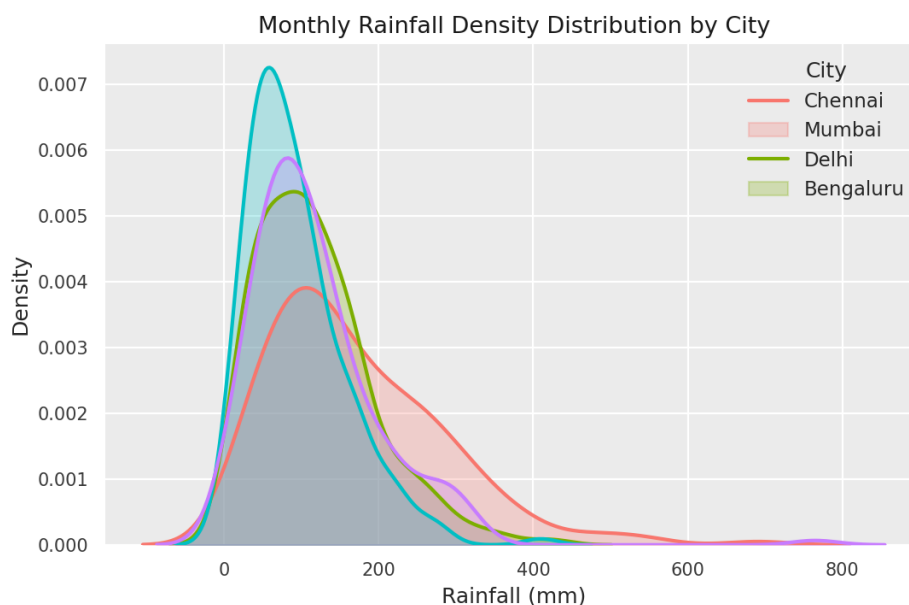


Figure: R (ggplot2) Output

### Comparative Insight

Both charts reveal right-skewed rainfall patterns, with some cities showing sharp, narrow peaks (drier, more consistent months) and others wider, flatter curves (higher variability). Visually, the Python version sits on a plain white grid with a Viridis-derived palette, while the R/ggplot2 version uses the grey panel background and ggplot2's default hue palette with slightly bolder outline strokes on each density curve.

## 5. Bank Loan Risk Assessment

### Task / Scenario

A financial institution has customer loan amounts, repayment history, income levels, and credit scores. Develop visualizations showing loan distributions and customer segments. Evaluate which visualization best supports loan approval decisions by highlighting risk patterns, skewness, and outliers, and recommend the visualization for a management dashboard.

### Selected Visualization & Justification

A Histogram (with density overlay) was chosen as the most decision-relevant visualization. It directly exposes the skewness of loan amounts — a large concentration of small-to-moderate loans with a long right tail of large, higher-risk loans — which is critical evidence for setting approval thresholds and risk-based pricing tiers.

### Python Implementation

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv('loan_data.csv') # column: LoanAmount

sns.histplot(df['LoanAmount'], bins=30, kde=True, color='teal')
plt.title('Distribution of Loan Amounts (Risk Skewness Check)')
plt.xlabel('Loan Amount ($)'); plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```

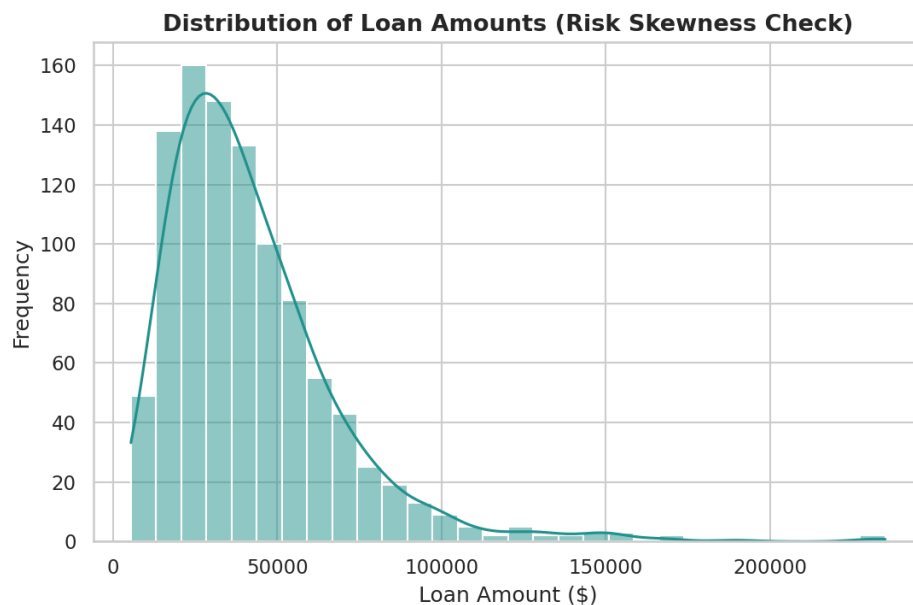


Figure: Python (Matplotlib/Seaborn) Output

### R Implementation

```
library(ggplot2)

df <- read.csv('loan_data.csv') # column: LoanAmount

ggplot(df, aes(x = LoanAmount)) +
  geom_histogram(bins = 30, fill = '#00BFC4', color = 'white') +
  labs(title = 'Distribution of Loan Amounts (Risk Skewness Check)',
       x = 'Loan Amount ($)', y = 'Frequency') +
  theme_gray(base_size = 12)
```

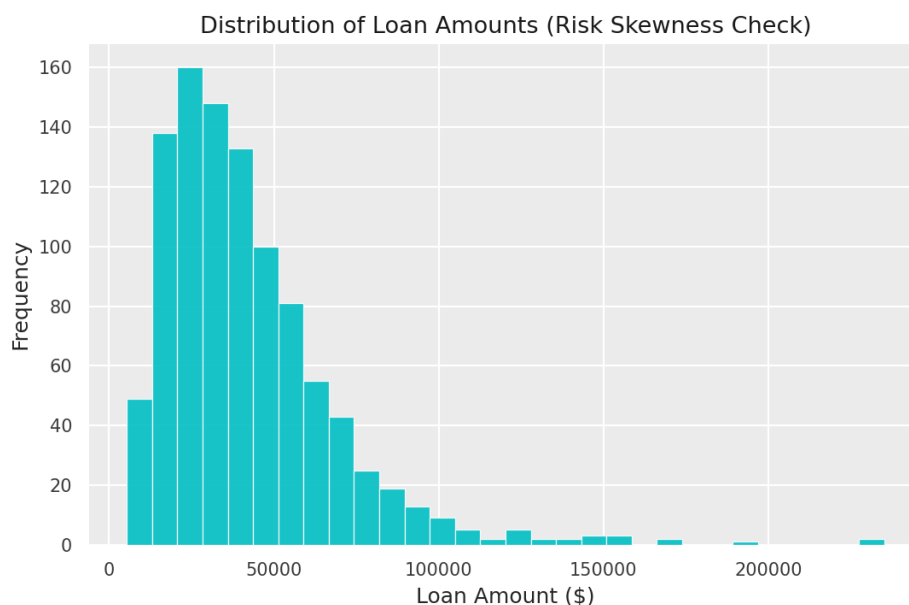


Figure: R (ggplot2) Output

### Comparative Insight

Both histograms confirm a strong positive (right) skew in loan amounts (skewness  $\approx 2.1$ ), meaning most customers borrow modest amounts while a small group of large loans pulls the tail — a key risk-concentration signal. The Python version overlays a smooth KDE curve on a white-gridded background, while the R/ggplot2 version renders plain bars with white borders on ggplot2's grey panel — giving the two outputs a clearly different visual finish despite showing the same underlying pattern.